Employee Database Design and Analysis

Objective

- To design a relational database for managing employee information.
- To perform data analysis using SQL queries.
- To apply concepts such as joins, aggregations, filtering, and grouping.
- To generate meaningful business insights from employee data.

Database Description

The project uses a structured relational dataset consisting of three main tables:

Employees

Column Name	Data Type	Description
employee_id	INT (PK)	Unique employee identifier
employee_name	VARCHAR(100)	Name of the employee
gender	CHAR(1)	Gender (M/F)
age	INT	Age of the employee
hire_date	DATE	Date of joining
designation	VARCHAR(100)	Job role/designation
department_id	INT (FK)	References Departments table
location_id	INT (FK)	References Location table
salary	INT	Employee salary

Departments

Column Name	Data Type	Description
department_id	INT (PK)	Unique department identifier
department_name	VARCHAR(100)	Department name

Location

Column Name	Data Type	Description
location_id	INT (PK)	Unique location identifier
location	VARCHAR(100)	Location name

Features Implemented

Data Definition & Manipulation

- Created structured tables for employees, departments, and locations
- Inserted multiple records into tables
- Updated missing values (designation field)

Query Operations

- **DISTINCT** – Retrieved unique salary values
- **Alias (AS)** – Renamed columns for better readability
- **WHERE Clause** – Filtered records based on conditions
- **ORDER BY** – Sorted data by department and salary
- **LIMIT** – Restricted output records

Aggregate Functions

- SUM – Total salary calculation
- MIN – Minimum age
- AVG – Average salary and age
- COUNT – Number of employees

Grouping & Filtering

- GROUP BY – Grouped data by department and location
- HAVING – Filtered grouped results

Joins

- INNER JOIN – Combined employee and department data
- LEFT JOIN – Included all departments with employee count
- RIGHT JOIN – Retrieved all locations with employee data

Key Insights

- Some departments have fewer employees, indicating possible understaffing
- Salary varies significantly across departments and roles
- Certain locations have higher average employee age
- Analysts have varied salary ranges depending on experience
- Missing designation data was identified and corrected

Conclusion

- This project demonstrates the effective use of SQL in designing and analyzing a relational employee database.
- It applies key concepts such as joins, aggregations, and data filtering to extract meaningful insights.
- The analysis supports better understanding of workforce distribution and enables data-driven decision-making.

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